

```
In [1]: csv_url = 'https://archive.ics.uci.edu/ml/machine-learning-databases
```

```
In [2]: import pandas as pd
```

```
In [3]: iris = pd.read_csv(csv_url, header = None)
```

```
In [4]: col_names = ['Sepal_Length', 'Sepal_Width', 'Petal_Length', 'Petal_Width
```

```
In [5]: iris = pd.read_csv(csv_url, names = col_names)
```

```
In [6]: df1=df=iris
```

```
In [7]: iris.head(8)
```

```
Out[7]:
```

	Sepal_Length	Sepal_Width	Petal_Length	Petal_Width	Species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa
5	5.4	3.9	1.7	0.4	Iris-setosa
6	4.6	3.4	1.4	0.3	Iris-setosa
7	5.0	3.4	1.5	0.2	Iris-setosa

```
In [8]: iris.tail()
```

```
Out[8]:
```

	Sepal_Length	Sepal_Width	Petal_Length	Petal_Width	Species
145	6.7	3.0	5.2	2.3	Iris-virginica
146	6.3	2.5	5.0	1.9	Iris-virginica
147	6.5	3.0	5.2	2.0	Iris-virginica
148	6.2	3.4	5.4	2.3	Iris-virginica
149	5.9	3.0	5.1	1.8	Iris-virginica

```
In [9]: iris.index
```

```
Out[9]: RangeIndex(start=0, stop=150, step=1)
```

```
In [10]: iris.columns
```

```
Out[10]: Index(['Sepal_Length', 'Sepal_Width', 'Petal_Length', 'Petal_Width',  
              'Species'],  
              dtype='object')
```

```
In [11]: iris.shape
```

```
Out[11]: (150, 5)
```

```
In [12]: iris.dtypes
```

```
Out[12]: Sepal_Length    float64  
Sepal_Width    float64  
Petal_Length    float64  
Petal_Width    float64  
Species        object  
dtype: object
```

```
In [13]: iris.describe()
```

```
Out[13]:
```

	Sepal_Length	Sepal_Width	Petal_Length	Petal_Width
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.054000	3.758667	1.198667
std	0.828066	0.433594	1.764420	0.763161
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

```
In [14]: iris.columns.values
```

```
Out[14]: array(['Sepal_Length', 'Sepal_Width', 'Petal_Length', 'Petal_Width',  
              'Species'], dtype=object)
```

```
In [15]: iris.iloc[5]
```

```
Out[15]: Sepal_Length    5.4  
Sepal_Width    3.9  
Petal_Length    1.7  
Petal_Width    0.4  
Species    Iris-setosa  
Name: 5, dtype: object
```

```
In [16]: iris[47:51]
```

```
Out[16]:
```

	Sepal_Length	Sepal_Width	Petal_Length	Petal_Width	Species
47	4.6	3.2	1.4	0.2	Iris-setosa
48	5.3	3.7	1.5	0.2	Iris-setosa
49	5.0	3.3	1.4	0.2	Iris-setosa
50	7.0	3.2	4.7	1.4	Iris-versicolor

```
In [17]: iris.loc[:, ["Sepal_Length", "Sepal_Width"]]
```

```
Out[17]:
```

	Sepal_Length	Sepal_Width
0	5.1	3.5
1	4.9	3.0
2	4.7	3.2
3	4.6	3.1
4	5.0	3.6
...	...	...
145	6.7	3.0
146	6.3	2.5
147	6.5	3.0
148	6.2	3.4
149	5.9	3.0

150 rows × 2 columns

```
In [18]: cols_2_4=iris.columns[2:4]
iris[cols_2_4]
```

```
Out[18]:
```

	Petal_Length	Petal_Width
0	1.4	0.2
1	1.4	0.2
2	1.3	0.2
3	1.5	0.2
4	1.4	0.2
...	...	...
145	5.2	2.3
146	5.0	1.9
147	5.2	2.0
148	5.4	2.3
149	5.1	1.8

150 rows × 2 columns

```
In [19]: iris.isnull().any()
```

```
Out[19]: Sepal_Length    False
Sepal_Width      False
Petal_Length     False
Petal_Width      False
Species          False
dtype: bool
```

```
In [20]: iris.isnull().sum()
```

```
Out[20]: Sepal_Length    0
Sepal_Width      0
Petal_Length     0
Petal_Width      0
Species          0
dtype: int64
```

```
In [21]: iris.dtypes
```

```
Out[21]: Sepal_Length    float64
Sepal_Width      float64
Petal_Length     float64
Petal_Width      float64
Species          object
dtype: object
```

```
In [22]: df=iris
df['petal Length(cm)']=iris['Petal_Length'].astype("int")
```

```
In [23]: df1=df
```

In [24]: df

Out[24]:

	Sepal_Length	Sepal_Width	Petal_Length	Petal_Width	Species	petal Length(cm)
0	5.1	3.5	1.4	0.2	Iris-setosa	1
1	4.9	3.0	1.4	0.2	Iris-setosa	1
2	4.7	3.2	1.3	0.2	Iris-setosa	1
3	4.6	3.1	1.5	0.2	Iris-setosa	1
4	5.0	3.6	1.4	0.2	Iris-setosa	1
...	...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	Iris-virginica	5
146	6.3	2.5	5.0	1.9	Iris-virginica	5
147	6.5	3.0	5.2	2.0	Iris-virginica	5
148	6.2	3.4	5.4	2.3	Iris-virginica	5
149	5.9	3.0	5.1	1.8	Iris-virginica	5

150 rows × 6 columns

In [25]:

```
from sklearn import preprocessing
min_max_scaler = preprocessing.MinMaxScaler()
```

In [26]: X=iris.iloc[:, :4]

In [27]: X

Out[27]:

	Sepal_Length	Sepal_Width	Petal_Length	Petal_Width
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2
2	4.7	3.2	1.3	0.2
3	4.6	3.1	1.5	0.2
4	5.0	3.6	1.4	0.2
...	...	...	...	...
145	6.7	3.0	5.2	2.3
146	6.3	2.5	5.0	1.9
147	6.5	3.0	5.2	2.0
148	6.2	3.4	5.4	2.3
149	5.9	3.0	5.1	1.8

150 rows × 4 columns

In [28]:

```
X_scaled = min_max_scaler.fit_transform(X)
```

```
In [29]: df_normalized = pd.DataFrame(X_scaled)
```

```
In [30]: df_normalized
```

```
Out[30]:
```

	0	1	2	3
0	0.222222	0.625000	0.067797	0.041667
1	0.166667	0.416667	0.067797	0.041667
2	0.111111	0.500000	0.050847	0.041667
3	0.083333	0.458333	0.084746	0.041667
4	0.194444	0.666667	0.067797	0.041667
...	...	...	...	...
145	0.666667	0.416667	0.711864	0.916667
146	0.555556	0.208333	0.677966	0.750000
147	0.611111	0.416667	0.711864	0.791667
148	0.527778	0.583333	0.745763	0.916667
149	0.444444	0.416667	0.694915	0.708333

150 rows × 4 columns

```
In [31]: df2=df
df2['Species'].unique()
```

```
Out[31]: array(['Iris-setosa', 'Iris-versicolor', 'Iris-virginica'], dtype=object)
```

```
In [32]: df_normalized = pd.DataFrame(X_scaled)
```

```
In [33]: df_normalized
```

```
Out[33]:
```

	0	1	2	3
0	0.222222	0.625000	0.067797	0.041667
1	0.166667	0.416667	0.067797	0.041667
2	0.111111	0.500000	0.050847	0.041667
3	0.083333	0.458333	0.084746	0.041667
4	0.194444	0.666667	0.067797	0.041667
...	...	...	...	...
145	0.666667	0.416667	0.711864	0.916667
146	0.555556	0.208333	0.677966	0.750000
147	0.611111	0.416667	0.711864	0.791667
148	0.527778	0.583333	0.745763	0.916667
149	0.444444	0.416667	0.694915	0.708333

150 rows × 4 columns

```
In [34]: df2=df  
df2['Species'].unique()
```

```
Out[34]: array(['Iris-setosa', 'Iris-versicolor', 'Iris-virginica'], dtype=object)
```

```
In [35]: from sklearn import preprocessing  
enc = preprocessing.OneHotEncoder()
```

```
In [36]: features_df=df2.drop(columns=['Species'])
```

```
In [37]: features_df
```

```
Out[37]:
```

	Sepal_Length	Sepal_Width	Petal_Length	Petal_Width	petal Length(cm)
0	5.1	3.5	1.4	0.2	1
1	4.9	3.0	1.4	0.2	1
2	4.7	3.2	1.3	0.2	1
3	4.6	3.1	1.5	0.2	1
4	5.0	3.6	1.4	0.2	1
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	5
146	6.3	2.5	5.0	1.9	5
147	6.5	3.0	5.2	2.0	5
148	6.2	3.4	5.4	2.3	5
149	5.9	3.0	5.1	1.8	5

150 rows × 5 columns

```
In [38]: enc_df=(enc.fit_transform(df2[['Species']])).toarray()
```

```
In [39]: enc_df = pd.DataFrame(enc_df, columns = ['Iris-Setosa', 'Iris-Versico
```

```
In [40]: df_encode = features_df.join(enc_df)
```

In [41]: df_encode

Out[41]:

	Sepal_Length	Sepal_Width	Petal_Length	Petal_Width	petal Length(cm)	Iris- Setosa	Iris- Versicolor
0	5.1	3.5	1.4	0.2	1	1.0	0.0
1	4.9	3.0	1.4	0.2	1	1.0	0.0
2	4.7	3.2	1.3	0.2	1	1.0	0.0
3	4.6	3.1	1.5	0.2	1	1.0	0.0
4	5.0	3.6	1.4	0.2	1	1.0	0.0
...	...	...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	5	0.0	0.0
146	6.3	2.5	5.0	1.9	5	0.0	0.0
147	6.5	3.0	5.2	2.0	5	0.0	0.0
148	6.2	3.4	5.4	2.3	5	0.0	0.0
149	5.9	3.0	5.1	1.8	5	0.0	0.0

150 rows × 8 columns

In []:

In []: